

DRUG INFORMATICS

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This course

- introduces the student to the **use of drug literatures** in the **promotion of safe, effective and rational drug therapy**
- helps the student to **develop the skills** for **retrieving, evaluating and communicating** drug information to **health care professionals** and the **patient**

- ❑ **At the end of the course, the student should be able to:**
 - Rapidly locate and evaluate drug information sources
 - Systematically manage and communicate drug information
 - Apply drug information knowledge for preparation and management of formularies, guidelines and bulletins
 - Provide drug information to health care professionals and patients on the rational use of drugs

In- Class activities & Assignments

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□ Includes:

- Using resources to locate answers for DI questions
- Developing search strategies for internet & 2ndary data bases
- Presenting journal Club and Apply critical literature evaluation skills using original research articles
- Evaluate medications for inclusion on formulary, Drug monograph evaluation
- Written drug information consult

Drug Information:

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□ Drug information

- is an area of pharmacy practice that deals with obtaining, managing, and evaluating information to prepare and disseminate it in a suitable format, wherever and whenever it is needed or in anticipation of need
- This area of practice is one of the oldest in clinical pharmacy
- Drug information skills are at the core of clinical pharmacy practice

For what it helps you?

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Information Services

- ▣ Provide concise, applicable, and timely responses to requests for information from members of the health care team.
 - Appropriately extract all background information necessary to respond to a request for drug information
 - Identify the most appropriate sources of information on the use of drugs in patients
 - Formulate a systematic, efficient, and thorough procedure for retrieving drug information.
 - Efficiently utilize both manual and computerized sources of drug information
 - Evaluate the quality of literature gathered
 - Effectively communicate responses to the requestor and/or team

Cont'd...helps

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- Drug information skills are required to deliver Pharmaceutical Care
- Students will be trained to develop the skills to obtain information from various literature and reference sources to answer drug information questions efficiently
- Techniques for researching and evaluating drug literature will be covered
- Emphasis will be placed on systemic approaches to formulation of responses utilizing both verbal and written communication skills

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- ❑ The provision of medication information is among the most fundamental responsibilities of pharmacists
 - ❑ *The information may be either patient-specific or relative to a group of patients, such as in the development of a therapeutic guideline, coordination of an adverse drug event reporting and analysis program, publication of an electronic newsletter, or updating a Web site.*

Definition: drug information

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- The term :drug information may have different meanings to different people depending on the context in which it is used
 - *Specific individual*: **Specialist/practitioner/pharmacist/provider**
 - *Place*: **Centre/service/practice**
 - *Activities and abilities of individuals*: **Functions/Skills**

- ❑ as *printed information in a reference* or verbalized by an individual that pertains to medications
 - ❑ *cost-effective medication selection & use, medication policy decisions (drug benefits), medication information resource selection, practice - related issues*
- ❑ The provision of unbiased, well-referenced, and critically evaluated information on drug related issues

Definitions....

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- Drug information:
 - Is the provision of written and/or verbal information or advice about drugs and drug therapy in response to a request from other HCPs, organizations, committees, patients or members of the public
- Drug information services:
 - Describes activities undertaken by pharmacists in providing information to optimize drug use, the term also includes but not limited to , the specialized services offered by a drug information center
- Drug information center
 - Refers to a facility specifically set aside for, and specializing in, provision of drug information

□ **Drug information Pharmacist**

- ▣ Refers to a pharmacist who have completed a course of training in drug information such as advanced courses for a specialty practice in drug information, and specializes in drug information

□ **Drug informatics:**

- ▣ refers to **the application of technology** in the delivery of drug information services
- ▣ denote the **electronic management of information**
- ▣ emphasizes the use of technology as an integral tool in effectively organizing, analyzing, managing, and communicating information on medication use in patients

□ **Drug information services, in turn, include:**

- ▣ responding to drug information inquiries, Conducting medication use evaluations and participating in medication quality assurance programs, such as; monitoring adverse drug reactions, drug and herbal product interactions, and medication errors

Definition....

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□ **Poison information**

- is a specialized area of medication information with the practitioner typically practicing in an accredited poison information center or an emergency room

□ **Poison control centers and Drug information controls**

- Both have a common goal “provide comprehensive, accurate, and timely information to their clients”
- Both used the information “to enhance the medical care of patients”
- Both have similar “information retrieval process and physical layouts”

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- Several differences between a traditional drug information center and poison control center: in poison center -
 - Public generate most consultations received in poison center
 - Public vs. health care professionals
 - Eighty eight percent of PCC calls came from public
 - Nine to 10 percent of DIC calls came from public
 - prepared to provide information on the management of any poison situation
 - nearly all requests for information to a poison control center are urgent, with an average response time of 5-minutes
 - A specialist in poison information requires expertise in clinical toxicology
 - Requires accreditation/national certification examination in most developed countries

- Relative to current practice, the term *medication information* is used in place of *drug information*:
 - ▣ to convey the management and use of information on medication therapy
 - ▣ To signify a broader role that all pharmacists take in information provision
- Drug informatics is used to describe the evolving roles of the medication information specialist
 - ▣ emphasizes the use of technology as an integral tool in effectively organizing, analyzing, and managing information on medication use in patients
 - ▣ describe how the role of the pharmacist has evolved in providing medication information

Goal: Improve patient outcomes

Tool: Clinical Drug Literature

Results: Beginning of the Clinical Pharmacy concept

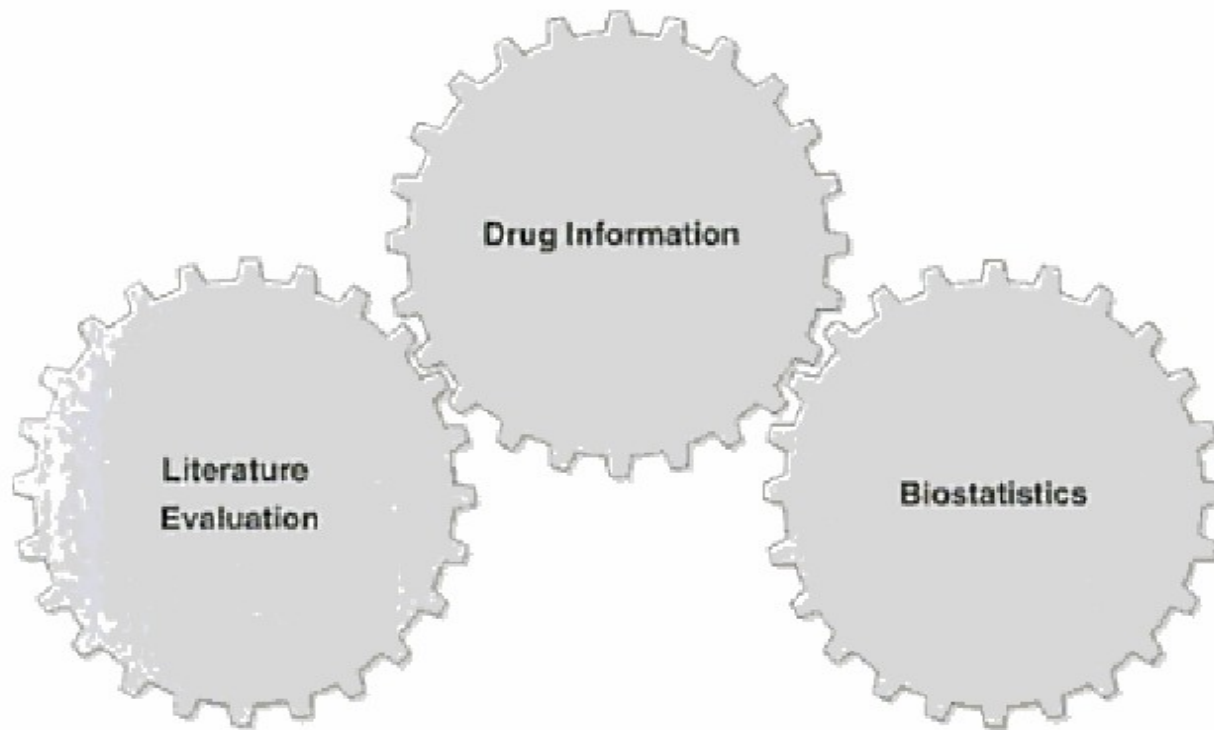


FIGURE  Integration of Drug Information, Biostatistics, and Literature Evaluation. Drug information, the provision of credible data on pharmacy-related issues, integrates biostatistics and literature evaluation.

Literature evaluation and biostatistics are key to critically appraising information before it is disseminated so valid recommendations regarding patient care can be made.

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THE BEGINNING.....

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History of DIC: the beginnings – Global aspect(USA)

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- 1864 – 1876: John Shaw Billings, a Civil War surgeon, the Surgeon General's library. In 1876 he declares as the National Library of Medicine (NLM)
- 1876-1879: Billings begins to catalog the contents of the library by overall medical subject addressed. The term Medical Subject Headings (MeSH) is used to describe the catalog headings
- 1900-1960: The number of available drugs and journal articles expands
 - ▣ It becomes more difficult for physicians to keep abreast of the medical literature. After World War II, a need is seen for information retrieval and not just cataloging
 - ▣ Development of such a system begins in the 1950s and uses Boolean search terms AND, OR, and NOT

- 1962: The first drug information center opens at the University of Kentucky Medical Center
- Function:
 - ▣ Source of selected, comprehensive drug information to evaluate and compare drugs
 - ▣ For staff physicians, dentists, nurses
 - ▣ Education of health professional students: Medicine, dentistry, nursing, and pharmacy
 - ▣ Stated goal: To influence pharmacy students in developing their role as drug consultants
- 1960s: Pharmacist-staffed drug information centers open around USA
 - ▣ Pharmacists are prohibited from giving patients any information about dispensed medications in early part of decade.
 - ▣ Around 1963, the practice act in Washington state is changed to permit pharmacists to provide information about legend drugs.

- The Medical Literature Analysis and Retrieval System (MEDLARS) is developed by the NLM;
 - ▣ this system uses computerized searches but information requests are submitted by telephone and results mailed to the requester
- 1971: the ability to transmit information over telephone lines is applied to MEDLARS;
 - ▣ The resulting database searching technology is called MEDLARS ON- LINE
 - ▣ Named as MEDLINE (for fee)
- 1973: a formal survey counts 54 pharmacist-operated drug information centers in USA

- 1975: A report issued by an external review board strongly recommends that all pharmacy students be trained to use drug information resources (USA)
- 1980s and early '90s: The number of drug information centers peaks.
 - ▣ In 1988, the National Center for Biotechnology Information (NCBI) database of biotechnology was created
- 1997: By congressional decree, Web access to MEDLINE information is made free to the public. The NLM partners with NCBI to develop PubMed,
 - a Web portal to access biomedical information from many databases, not just MEDLINE
 - Web sites with drug and disease information of varying quality are burgeoning
- The American Society of Health-System Pharmacists (ASHP) provides residency accreditation in drug information

History of DI services: Ethiopia

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- Academia (AAU), some Hospitals & NGO started activities in relation to providing drug information , esp. in r/n to ARV drugs
- EFDA, then DACA
- AAU , School of pharmacy launched DIC
- Mekele and Gondar University teaching hospitals: 2010
- St. Paul Hospital: 2010
- Alert Hospital 2011
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Definitions: *Drug Information Service*

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- Describes the activities undertaken by pharmacists in providing information to optimize drug use.
- It is those services which cover the activities of specially trained individual to provide accurate, unbiased, factual information in response to patient oriented drug problems that are received from health care professionals

Medication Information Services

✓ Supporting clinical services with medication information	✓ Coordinating formulary management initiatives
✓ Answering questions regarding medications	✓ Developing criteria/guidelines for medication use
✓ Coordinating pharmacy and therapeutics committee activity	✓ Analyzing the clinical and economic impact of drug policy decisions
✓ Developing medication use policies	✓ Managing medication usage evaluation/medication use evaluation
✓ Publishing or editing information on appropriate medication use through newsletters, journal columns, Web sites, etc.	✓ Providing education (e.g., in-services, classes, experiential education) for health professionals, students, and consumers
✓ Managing investigational medication use (e.g., institutional review board activities, information for practitioners)	✓ Coordinating of adverse drug event reporting and analysis programs, e.g., adverse medication reactions
✓ Providing poison information	

Medication information skills

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- Assess available information and gather situational data needed to characterize question or issue
- Formulate appropriate question(s)
- Use a systematic approach to find needed information
- Evaluate information critically for validity and applicability
- Develop, organize, and summarize response for question or issue
- Communicate clearly when speaking or writing, considering the audience level
- Anticipate other information needs

- Drug information specialists are pharmacists whose primary responsibility is the provision of drug information
 - ▣ The role has changed from an individual who specifically answers questions to one who focuses on the development of medication policies and provides information on complex medication information questions

The Specialists.....

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- Drug information specialists work in a variety of settings, each with its own unique scope of practice.
 - ▣ Academic drug information centers
 - ▣ Proprietary and generic drug manufacturers
 - ▣ Contract Drug Information Center (Fee-for-Service)
 - providing answers to drug information requests
 - preparation of new drug evaluation monographs
 - formulary drug class reviews
 - development of MUE criteria
 - providing journal reprints
 - Pharmacoeconomics evaluations
 - writing a pharmacotherapy newsletter, and providing continuing education programming.
- Health Maintenance Organizations /Managed Care Organizations
- Poison Control

- Drug information is both a body of data and information about medications and a set of skills and tools that provide pharmacy professionals with the ability
 - ▣ to find, access, understand, interpret, apply and communicate information and acquire knowledge.
- The body of facts and information pertaining to medications is generally referred to as “the drug literature”

DRUG INFORMATION - What is it?

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1. Identifying the ultimate question
2. Locating
 - a. Resources knowledge
 - b. Techniques of searching
3. Analyzing
 - a. Statistics
 - b. Literature evaluation
4. Applying information to the question
Pharmacotherapy/Pharmacology
5. Communicating information

- Health care providers and patients are continually look to pharmacists for unbiased , evidence based , accurate information regarding drug therapies and pharmaceutical related issues
- Question: Why are drug information question asked?
 - ▣ The answer is simple.
 - Questions are asked b/c medicine is constantly evolving.
 - Everything is not known or fully understood and likely never will be.
 - We are always searching for a better understanding of health and disease and for truth in the universe

Case scenarios :

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- A 35 year old woman complains of a three day history of cough productive of moderate amounts of greenish sputum. Her illness started six days ago with a sore throat was followed by nasal congestion, a brief episode of hoarseness, and then the cough she currently has. The nasal congestion, sore throat, and hoarseness have all resolved but the cough seems to be getting worse. She feels a mild tightness substernally and has occasional wheezes, but she denies any shortness of breath, fever, or pleuritic chest pain.

Cont'd....

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But she has no history of asthma and has never smoked cigarettes. On physical exam you find only a few scattered rhonchi bilaterally. Otherwise the physical exam is entirely normal. You make a diagnosis of acute bronchitis and the patient says she has usually been prescribed an antibiotic for this type of problem in the past. Although you have prescribed antibiotics for similar patients in the past, you have recently heard that antibiotics may not be indicated in all cases of bronchitis. You wish to find out whether an antibiotic would be appropriate for this patient.

Functions of DI Services

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1. Responding to DI questions
2. Drugs and Therapeutics Committee (DTC) and Formulary Support
3. Drug Utilization Review(DUR)/Drug Utilization Evaluation
4. Adverse Drug Reaction (ADR) Reporting
5. Patient Counseling
6. Continuing Education (CE) Programs
7. Investigational Drug Studies
8. Research
9. Drug Information Education

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THE EVOLUTION..... OF DRUG INFORMATION PRACTICE

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Factors Influencing the Evolution of the Pharmacist's Role as a Medication Information Provider

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1. Changing philosophy of practice
 2. Prevention of adverse drug events (ADEs)
 - 1999 IOM report, *To Err is Human: Building a Safer Health Care System*
 - ~7000 American deaths per year
 - Cost of drug - related morbidity and mortality: ~ U.S. \$77 billion/yr
 - ADE would cost lives and resources
 3. Growth of information technology
- Immediate information on
- ✓ new medication becomes commercially available
 - ✓ Withdrawn from the market for safety reasons
 - ✓ Data from a new study are released (impact on how common ailments are treated

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Influences...

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4. Changes in the health care environment with a focus on Evidence – Based Medicine (**EBM**) and Evaluation of **outcomes**
 - EBM: An approach to **practice** and **teaching** that **integrates** current **clinical research evidence** with *pathophysiologic rationale, professional expertise, and patient preferences* to **make decisions** for a population
5. Sophistication of medication therapy
 - Difficult for physicians and other health professionals to **keep up** with all of the **developments** in **medication therapy** and information required for their practice
6. More knowledgeable patient (self-care movement)
 - Health care cost
7. Improved accessibility of health information

Summary & Direction for the future

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- All pharmacists must be effective medication information providers
- An effective provider
 - ▣ Perceives, Assesses, and Evaluates medication information needs and Retrieves, Evaluates, Communicates, and Applies data from the published literature and other sources as an integral component of patient care
- If the profession is to be successful in accepting patient care responsibilities, **all pharmacists must have a certain minimum level of skill to survive** in the changing practice environment
- Developing the skills of an effective medication information provider is the foundation for the pharmacist to be a **lifelong learner and problem solver**
- **(Tool: Medical Literature)**

Summary & Direction for the future

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- Medication information skills is required in all practice settings (Generalist or a specialist practitioner)
- There is still the need for the practitioner to have support from drug information centers to meet
 - ▣ **Special information needs**
 - ▣ **to serve as a resource on effective medication use**
 - ▣ **to assist pharmacy practitioners as well as others in solving medication therapy situations**
- Individuals with special training as medication information specialists will still be needed to operate the centers and to provide leadership in the area of **Drug Informatics, Institutional drug policy, Poison control, Pharmaceutical industry, and in Academia**